

Ibn Tofail University

Analysis IV

Problem Set 2

Exercise 1:

Study the nature of convergence of each of the following sequences of functions:

1. $f_n(x) = \frac{1}{1+n(nx-1)^2}$, for $x \in I = [0, 1]$.
2. $f_n(x) = \begin{cases} (n-1)x & \text{if } x \in [0, \frac{1}{n}] , \\ 1-x & \text{if } x \in [\frac{1}{n}, 1] . \end{cases}$
3. $f_n(x) = e^{-nx} \cos x^2$, for $x > 0$.

Correction

Exercise 2:

Consider the sequence of functions $f_n : [-1, 1] \rightarrow \mathbb{R}$, defined by

$$x \mapsto f_n(x) = e^{-nx^2} \sin nx + \sqrt{1-x^2}, \quad n \in \mathbb{N}.$$

1. Show that the sequence of functions (f_n) converges pointwise on $I = [-1, 1]$ to a function f , which you will determine.
2. Show that (f_n) converges uniformly to f on $[\omega, 1]$, where ω is a positive constant.
3. Show that (f_n) does not converge uniformly to f on $[0, 1]$.

Correction

Exercise 3:

Determine the domain of convergence D of each series:

1. $\sum_{n>0} \frac{\cos nx}{n!}$.
2. $\sum_{n>1} \frac{1}{(z-n)^2}$.
3. $\sum_{n>1} \left| \sin \frac{x}{2^n} - \tan \frac{x}{2^n} \right|^{\frac{1}{2}}$.
4. $\sum_{n>1} \frac{n!(x-3)^{2n}}{n^{n+1}}$.

Correction**Exercise 4:**

Consider the following series of functions:

$$g_n(x) = \frac{\sin x}{x^2 + 1} + \frac{\sin 2x}{x^2 + 2^2} + \dots + \frac{\sin nx}{x^2 + n^2} + \dots$$

1. Determine the domain of convergence D .
2. Study the uniform convergence on D .

Correction**Exercise 5:**

Consider the series of functions with general term:

$$f_n(x) = \frac{nx}{1 + n^2x^2} - \frac{nx + x}{1 + n^2x^2 + 2n^2x + x^2}$$

1. Determine the domain of convergence D .
2. Calculate the sum of the series.
3. What is the nature of convergence of this series on D ?
4. Determine the nature of convergence of this series on $I =]0, +\infty[$ and $I = [a, +\infty[$ with $a > 0$.

Correction**Exercise 6:**

Consider the series of functions with general term: $f_n(x) = (\sin x)^\alpha (\cos x)^n$.

1. Determine the sum and domain of convergence of this series.
2. Show that we have uniform convergence on any compact $[a, b]$ included in $]0, \frac{\pi}{2}[$.

Correction